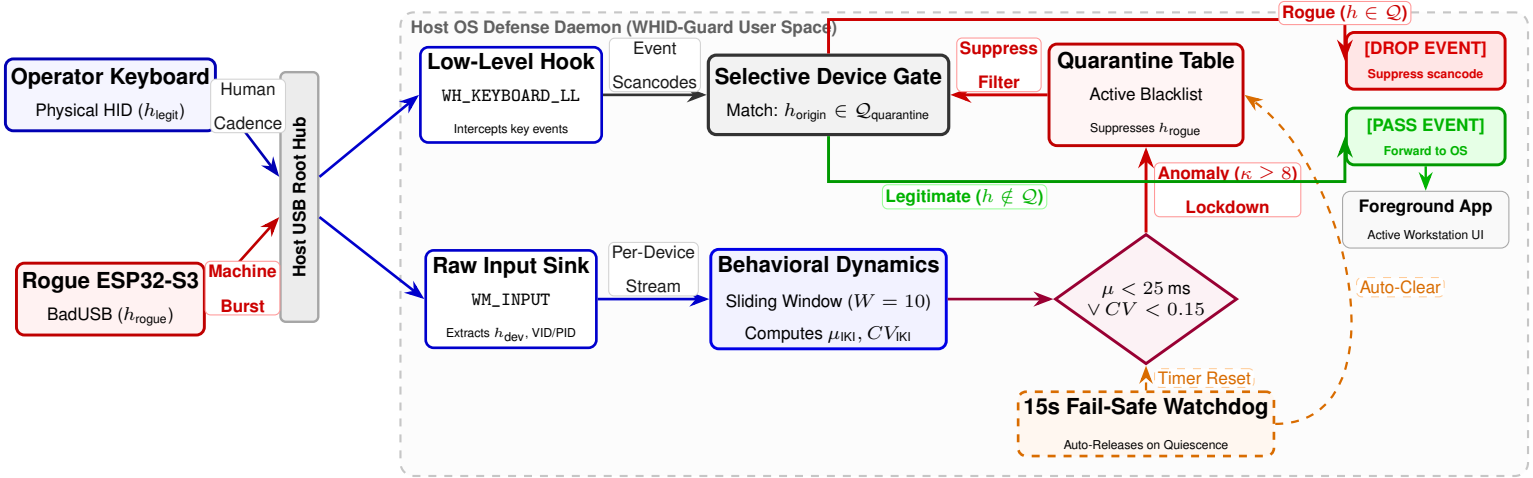

Host OS Defense Daemon (WHID-Guard User Space)
Operator Keyboard
Physical HID (h_{legit})
Human Cadence
Rogue ESP32-S3
BadUSB (h_{rogue})
Machine Burst
Host USB Root Hub
Low-Level Hook
WH_KEYBOARD_LL
Intercepts key events
Event Scancodes
Raw Input Sink
WM_INPUT
Extracts h_{dev}, VID/PID
Per-Device Stream
Selective Device Gate
Match: $h_{\text{origin}} \in \mathcal{Q}_{\text{quarantine}}$
Suppress Filter
Quarantine Table
Active Blacklist
Suppresses h_{rogue}
Rogue ($h \in \mathcal{Q}$)
[DROP EVENT]
Suppress scancode
[PASS EVENT]
Forward to OS
Foreground App
Active Workstation UI
Legitimate ($h \notin \mathcal{Q}$)
Anomaly ($\kappa > 8$)
Lockdown
Behavioral Dynamics
Sliding Window ($W = 10$)
Computes $\mu_{\text{IKI}}, CV_{\text{IKI}}$
$\mu < 25$ ms $\vee$ $CV < 0.15$
Auto-Clear
Timer Reset
15s Fail-Safe Watchdog
Auto-Releases on Quiescence